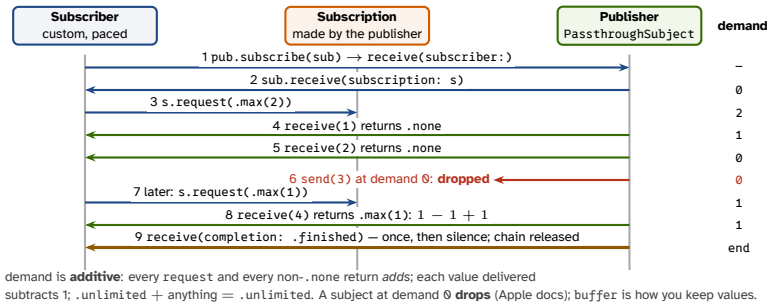
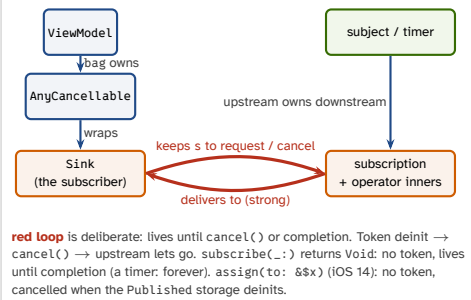


**In one line:** A **Publisher** is a recipe; `subscribe(_:)` makes it create a **Subscription** and hand it to the **Subscriber**, which **requests** a **demand**. The publisher may send **at most that many** values, then **one** completion. Demand is Combine's backpressure — but sink/assign ask `.unlimited`, so a push source that outruns its consumer must **drop or buffer**.

#### One subscription, demand counted step by step



#### Who keeps the chain alive



### The three protocols

- Publisher** <Output, Failure: Error>: one requirement, `receive(subscription:)`; you call `subscribe(_:)`. Operators are mostly **structs** wrapping upstream (`Publishers.Map<Upstream, T>`; `Share` is a class).
- Subscriber** (Input/Failure must match): `receive(subscription:)`, `receive(_:)` → `Subscribers.Demand` ("how many more"), `receive(completion:)` with `.finished` | `.failure(e)`.
- Subscription**: `Cancellable`: `request(_:)` + `cancel()`. `Cancel` is terminal and **silent**: no completion event follows.
- Demand** `.none` / `.max(n)` / `.unlimited`: values  $\leq$  demand,  $\leq 1$  completion, calls **serial**.

#### Example — a paced subscriber (real backpressure)

```
final class OneAtATime: Subscriber {
    typealias Input = Data; typealias Failure = Never
    private var sub: Subscription? // needed to ask again
    func receive(subscription s: Subscription) {
        sub = s; s.request(.max(1)) // demand 1
    }
    func receive(_ d: Data) -> Subscribers.Demand {
        upload(d) { [weak self] in self?.sub?.request(.max(1)) }
        return .none // pause until upload ends
    }
    func receive(completion: Subscribers.Completion<Never>) {
        sub = nil // drop the loop
    }
}
frames.buffer(size: 20, prefetch: .keepFull, whenFull: .dropOldest)
    .subscribe(OneAtATime()) // Void: no AnyCancellable
```

### Backpressure in practice

- `buffer(size:prefetch:whenFull:)` absorbs bursts. Prefetch `.keepFull` (fill at subscription, stay full) or `.byRequest`; when full `.dropNewest` / `.dropOldest` / `.customError { ... }` (fails the stream).
- Demand-shaping operators: `flatMap(maxPublishers: .max(n))` ( $\leq n$  inner subscriptions), `collect(n)` (batches), `.values` (1 per loop turn); rate-cutting: `throttle`, `debounce`.
- See it:** `.print("tag")` logs every request, value, completion, cancel; `handleEvents(receiveRequest:)` hooks them.

### Subjects in depth

- Classes conforming to `Subject`: `send(_:)`, `send(completion:)`, `send(subscription:)` — so a subject can *subscribe* to a publisher (`pub.subscribe(subject)`) and become one multicast point.
- `PassthroughSubject`: no initial value, no buffer — an event sent in `init`, before the view subscribed, is gone.
- `CurrentValueSubject`: `.value` readable synchronously; setting it *is* send; a new subscriber gets the current value first.
- After `send(completion:)` a subject is **finished for good**: later sends ignored, late subscribers get only the completion.
- `@Published's $x` acts like a `CurrentValueSubject<T, Never>` that fires on **willSet**.

### Schedulers — where code runs

- `Scheduler = now + schedule(...)` (now / after / repeating): `DispatchQueue`, `RunLoop`, `OperationQueue`, `ImmediateScheduler`.
- No receive(on:)** → delivered on the thread that emitted: `dataTaskPublisher = URLSession's background queue`, `@Published` = the setter's thread, `NotificationCenter.publisher` = the poster's thread.
- `receive(on:)` re-schedules every downstream event, values and completion. On `DispatchQueue.main` it is an **async hop** even when already on main — the first value lands a run-loop turn later.
- `subscribe(on:)` moves **subscribe, request and cancel** upstream (docs). A *synchronous* publisher (`Just`, `[1, 2].publisher`) emits *inside* request, so its values move too; `URLSession` still emits on its own queue.
- Time operators take a scheduler and **deliver on it**: `debounce`, `throttle`, `delay`, `timeout`, `measureInterval`. Inject it → testable.

### Traps

- Discarded token:** `Just(1).sink{...}` "works" (it finishes synchronously before the token dies); the same line on a network publisher never fires.
- Returning `.none` and never calling request again stalls the stream — no error, no completion, nothing.
- `.unlimited` + fast producer + `receive(on: .main)` = main queue flooded. Cut upstream: `throttle`, `collect`, `buffer(...dropOldest)`.
- send into one subject from several threads at once breaks the serial rule — funnel through one queue or actor.
- Cycle is not only in sink: any stored operator closure (`map`, `flatMap`, `handleEvents`) that captures `self` closes the loop too.

### Remember

**Subscribe** → **request** →  $\leq$  **demand values** → **one ending**. Push can't wait: drop or buffer. Upstream owns downstream; the **token** owns the loop.